

Assignment 19.2 Ai Assisted Coding

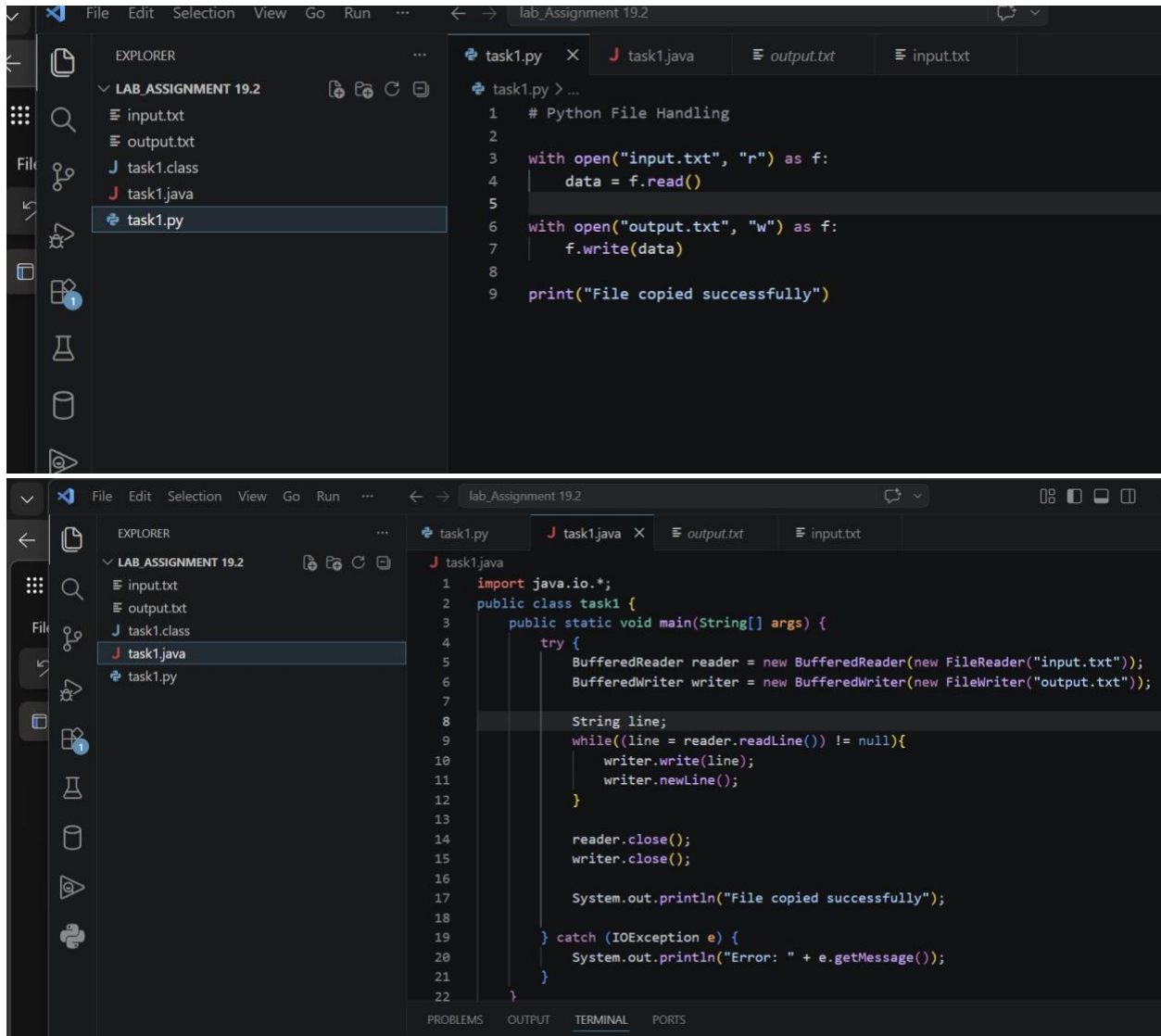
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Task 1: Python to Java Conversion (File Handling) Prompt:

Convert a Python file handling program into Java.

Code:



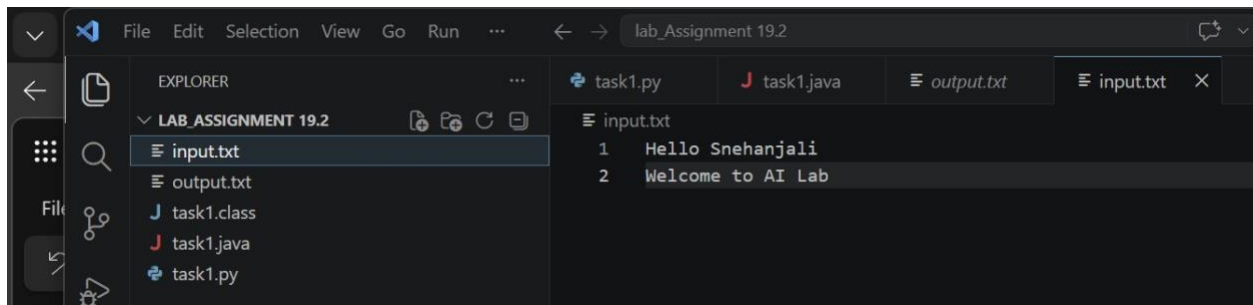
The image displays two screenshots of an IDE (Visual Studio Code) showing the conversion of a Python file handling program into Java.

Top Screenshot: The IDE shows the Python file `task1.py` selected in the Explorer. The code in `task1.py` is as follows:

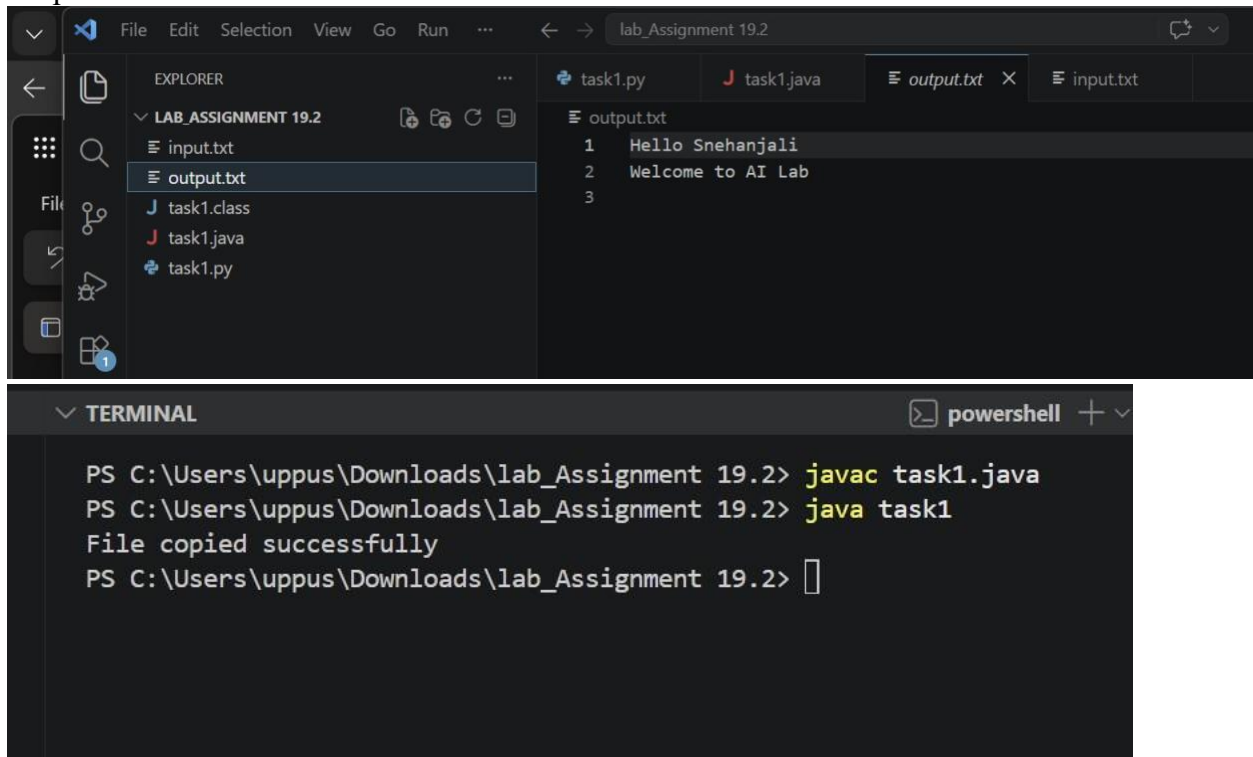
```
1 # Python File Handling
2
3 with open("input.txt", "r") as f:
4     data = f.read()
5
6 with open("output.txt", "w") as f:
7     f.write(data)
8
9 print("File copied successfully")
```

Bottom Screenshot: The IDE shows the Java file `task1.java` selected in the Explorer. The code in `task1.java` is as follows:

```
1 import java.io.*;
2 public class task1 {
3     public static void main(String[] args) {
4         try {
5             BufferedReader reader = new BufferedReader(new FileReader("input.txt"));
6             BufferedWriter writer = new BufferedWriter(new FileWriter("output.txt"));
7
8             String line;
9             while((line = reader.readLine()) != null){
10                 writer.write(line);
11                 writer.newLine();
12             }
13
14             reader.close();
15             writer.close();
16
17             System.out.println("File copied successfully");
18
19         } catch (IOException e) {
20             System.out.println("Error: " + e.getMessage());
21         }
22     }
23 }
```



Output:



Explanation:

The Python program reads data from a file and writes it to another file using open() function.

The Java program uses BufferedReader and BufferedWriter for file handling.

Exception handling is required in Java.

Both programs perform the same operation and produce identical output.

Task2: C++ to Python Conversion (Sorting)

Prompt:

Convert a C++ sorting program into Python.

Code:

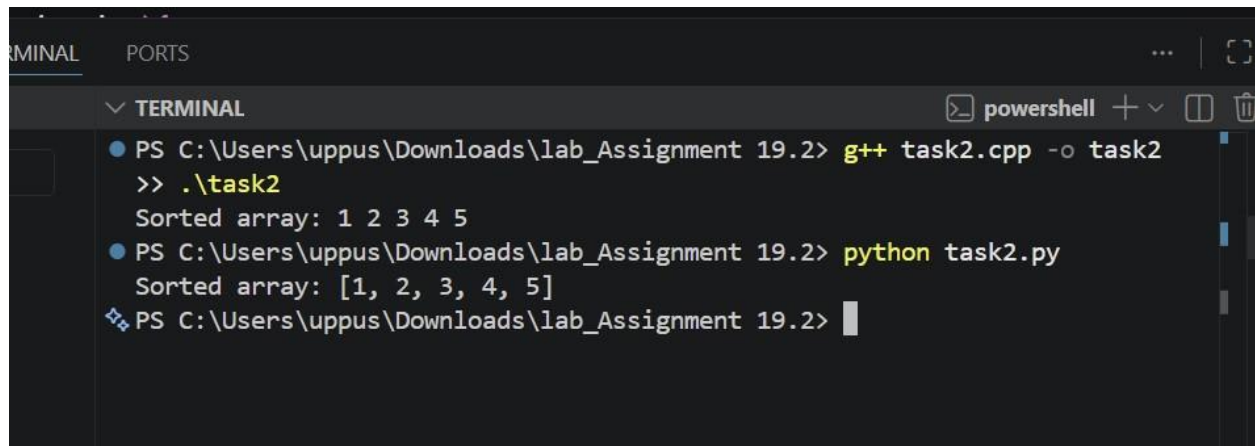
The screenshot shows the Visual Studio Code editor interface. The Explorer sidebar on the left displays a project named 'LAB_ASSIGNMENT 19.2' containing files: 'input.txt', 'output.txt', 'task1.class', 'task1.java', 'task1.py', 'task2.cpp' (selected), 'task2.exe', and 'task2.py'. The main editor window shows the code for 'task2.cpp'. The code implements a bubble sort algorithm in C++:

```
task2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int arr[5] = {5, 2, 4, 1, 3};
6
7      for(int i=0;i<5;i++){
8          for(int j=0;j<5-i-1;j++){
9              if(arr[j] > arr[j+1]){
10                 swap(arr[j], arr[j+1]);
11             }
12         }
13     }
14
15     cout << "Sorted array: ";
16     for(int i=0;i<5;i++)
17         cout << arr[i] << " ";
18
19     return 0;
20 }
```

The screenshot shows the Visual Studio Code editor interface with the same project structure as the first image. The Explorer sidebar now has 'task2.py' selected. The main editor window shows the code for 'task2.py', which implements a bubble sort algorithm in Python:

```
task2.py > ...
1  arr = [5, 2, 4, 1, 3]
2
3  for i in range(len(arr)):
4      for j in range(len(arr) - i - 1):
5          if arr[j] > arr[j + 1]:
6              temp = arr[j]
7              arr[j] = arr[j + 1]
8              arr[j + 1] = temp
9
10 print("Sorted array:", arr)
```

Output:



A terminal window titled 'TERMINAL' with a 'powershell' icon. It shows three commands and their outputs:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task2.cpp -o task2
>> .\task2
Sorted array: 1 2 3 4 5
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task2.py
Sorted array: [1, 2, 3, 4, 5]
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The C++ program uses Bubble Sort to sort an array.

-->The Python program follows the same logic.

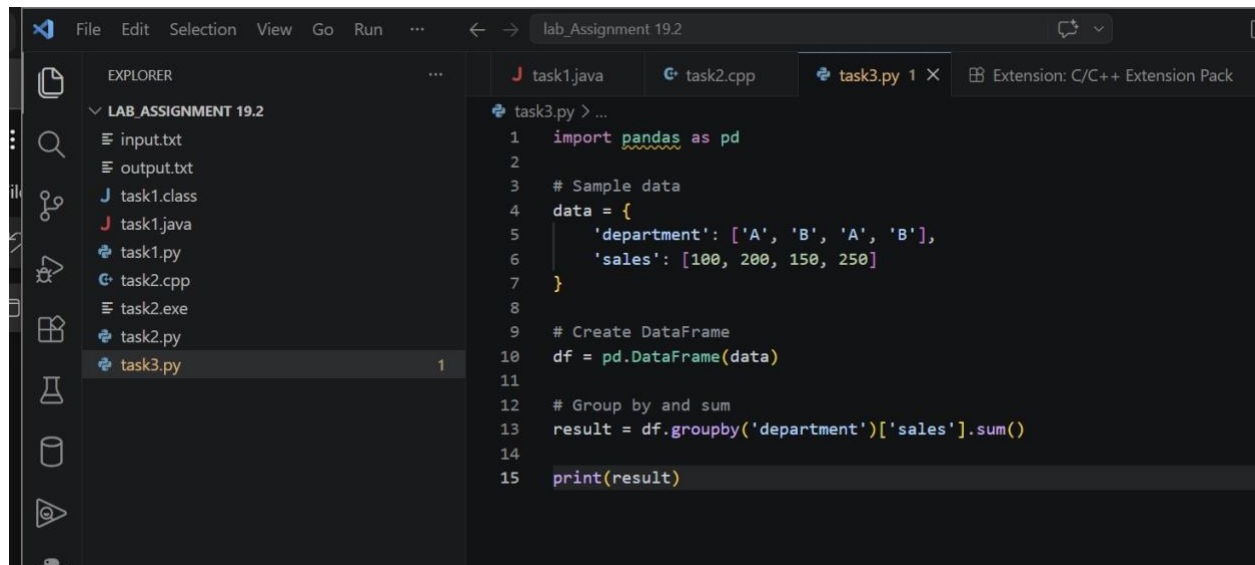
-->Both programs produce the same sorted result.

Task 3: SQL to Pandas Conversion

Prompt:

Convert a SQL query with GROUP BY into Pandas.

Code:



A Visual Studio Code editor window showing a file named 'task3.py'. The Explorer sidebar on the left shows a project named 'LAB_ASSIGNMENT 19.2' with files: 'input.txt', 'output.txt', 'task1.class', 'task1.java', 'task1.py', 'task2.cpp', 'task2.exe', 'task2.py', and 'task3.py' (selected). The main editor area shows the following Python code:

```
1 import pandas as pd
2
3 # Sample data
4 data = {
5     'department': ['A', 'B', 'A', 'B'],
6     'sales': [100, 200, 150, 250]
7 }
8
9 # Create DataFrame
10 df = pd.DataFrame(data)
11
12 # Group by and sum
13 result = df.groupby('department')['sales'].sum()
14
15 print(result)
```

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task3.py
department
A    250
B    450
Name: sales, dtype: int64
PS C:\Users\uppus\Downloads\lab_Assignment 19.2>
```

Explanation:

-->The SQL query groups data by department and calculates total sales.

-->The Pandas code uses `groupby()` and `sum()` to achieve the same result.

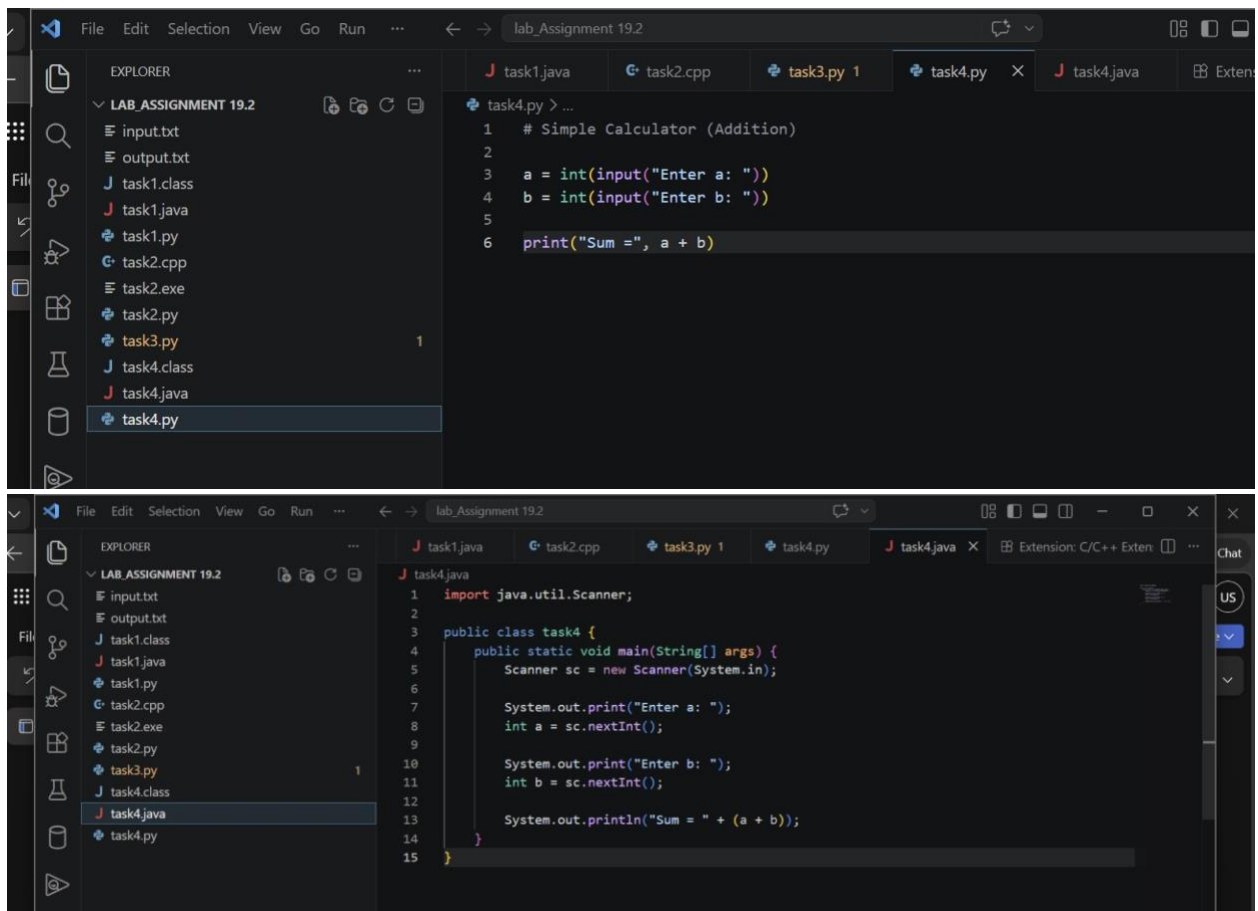
-->Both produce identical outputs.

Task 4: Real-Time Code Conversion (Python to Java)

Prompt:

Convert a Python calculator program into Java.

Code:



Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task4.py
Enter a: 3
Enter b: 5
Sum = 8
```

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> javac task4.java
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> java task4
Enter a: 3
Enter b: 5
Sum = 8
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program performs addition using simple input and print statements.

-->The Java program uses Scanner for input and follows class-based structure.

-->Both programs perform the same operation and produce identical output.

Task5: Python to C++ Conversion (Even/Odd)

Prompt:

Convert a Python program to C++ to check even or odd number.

Code:

The image shows two screenshots of the Visual Studio Code editor. The top screenshot displays the Python code for checking even or odd numbers. The bottom screenshot displays the C++ code for the same task.

Top Screenshot (Python Code):

```
task5.py > ...
1  # Check Even or Odd
2
3  n = int(input("Enter number: "))
4
5  if n % 2 == 0:
6      print("Even")
7  else:
8      print("Odd")
```

Bottom Screenshot (C++ Code):

```
task5.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter number: ";
7      cin >> n;
8
9      if(n % 2 == 0)
10         cout << "Even";
11     else
12         cout << "Odd";
13
14     return 0;
15 }
```

Output:

```
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> python task5.py
Enter number: 3
Odd
● PS C:\Users\uppus\Downloads\lab_Assignment 19.2> g++ task5.cpp -o task5
>> .\task5
Enter number: 3
○ Odd
PS C:\Users\uppus\Downloads\lab_Assignment 19.2> 
```

Explanation:

-->The Python program checks whether a number is even or odd using modulus operator.

-->The C++ program follows the same logic using if-else condition.

-->Both programs produce correct results for all test cases.